

ABDULRAHMAN

Machine Learning Researcher | Backend Engineer | Python Developer

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PROFILE

Computer Science undergraduate and researcher building systems end to end, from reinforcement learning agents trained in physics simulation to production backend and automation pipelines. Trained a humanoid robot to balance, resist external pushes, and walk using a custom PPO policy built from scratch, and authored an independent empirical study on zero knowledge machine learning proof cost. Combines a strong foundation in Python and machine learning with hands on ERP integration experience from real software house environments.

RESEARCH

Empirical Study on ReLU Count and Zero Knowledge Proof Cost

Independent Research

- Investigated how ReLU activation count predicts Zero Knowledge SNARK proof generation cost under the ezkl / Halo2 proving stack, training five parameter matched MNIST model variants in PyTorch.
- Found strong correlation between ReLU count and both prove time and proof size (Pearson r of 0.90 and 0.91), with model accuracy held stable across variants.
- Published with DOI: 10.5281/zenodo.21266805

Humanoid Robot: Reinforcement Learning in Physics Simulation

Personal Research Project

- Built and trained a humanoid robot from scratch using a custom MLP policy in PyBullet with Stable Baselines3 PPO, across three phases: balance, resisting pushes of up to 100N, and walking.
- Designed the full reward shaping and training pipeline independently, without a pre-built RL environment template.

PROJECTS

Gesture Controlled Robotic Hand

Personal Project

- Built a real time robotic hand that mirrors human finger gestures, translating computer vision hand landmark data into low latency servo motor commands.

ERP Modules: Auth, Attendance, Payroll

Personal Project

- Designed a microservices based ERP system (FastAPI, React, PostgreSQL) with independent authentication, attendance, and payroll services.

Real Estate Plot Management System / PayTrack

Client Projects

- Delivered a map based real estate plot management web app and a desktop installment tracking application with dashboards for payment schedules, balances, and overdue alerts, for paying clients.

EXPERIENCE

Full Stack Developer Intern, Netsol Technologies Pakistan

6 Weeks

- Worked on microservices based ERP modules and built authentication systems for enterprise facing modules within a large scale ERP platform.

Python Developer Intern, EUSOL Software House

6 Weeks

- Developed Python based automation scripts for reporting and ML assisted data processing, reducing manual effort and improving accuracy across departments.

EDUCATION

Bachelor of Computer Science, PUCIT

Present

CGPA: 3.69 / 4.0

Intermediate, Punjab Group of Colleges

1121 / 1200

Matriculation, Unique Group of High School

1065 / 1100

SKILLS

Languages: Python, C++, JavaScript, HTML/CSS

ML & Robotics: PyTorch, Stable Baselines3, PPO, PyBullet, NumPy, Pandas, scikit-learn, Neural Networks, Computer Vision

Backend & Web: FastAPI, Flask, Django, MERN Stack, Microservices, n8n